

NEUROGENESIS: From Fixed Models to Self-Evolving Intelligence

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Genre: Self-learning Intelligence system

Abstract:

The traditional AI models are trained on historical data and the data-driven models are often unable to cope with the changing data with the time. We propose a framework **NEUROGENESIS** based on HELIX (Hierarchical Evolutionary Learning & Intelligent eXaptation), that can constantly assess its performance, detect and discover changes in the input data and automatically decide which of the following adaptation options to pursue: retraining, incremental learning, or ensemble expansion. **NEUROGENESIS** is inspired by DNA—when the critical drift detection component is joined with a memory of previous knowledge to minimise forgetting. To evaluate HELIX we use the Allen Institute's Developing Mouse Visual Cortex single-cell dataset, incorporating developmental stages as successive data spaces. We want to demonstrate that AI can "learn" more as the data changes but retain the previous knowledge it acquired; and do this with few inputs from humans.

keywords: Memory Based learning, Streaming Machine Learning, Concept Drift,

I. INTRODUCTION:

Machine learning/Deep learning models trained on historical data, performance can decrease over time due to shifts in distribution of future data points. This makes algorithms less reliable in financial, biological or social media domains where data evolves constantly at the expense of expensive and/or time-consuming retraining of the models.

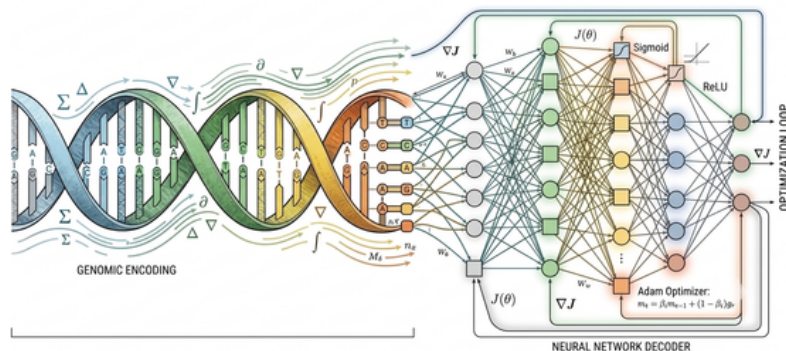
The paradigm is reinvented here and we presents the framework , **NEUROGENESIS** of an "artificial genome" that enables digital systems to adapt, repair and evolve like natural organisms. Not one specific algorithm, but three fundamental attributes define this architecture:

- **Self-awareness** - Ongoing assessment of model's results and data stream.
- **Self-Update** – Changes are detected and the most suitable change strategy (retrain, partial fit or ensemble) is selected automatically.
- **Self preservation** – Retaining previous knowledge by remembering the more important information first with the helps of memory buffer with importance weight replay prevents catastrophic forgetting.

The HELIX system is not a futuristic AGI, it is a working prototype. We show this on the Allen Institute developmental brain data set in which gene expression patterns vary between five different stages of development (embryonic to adulthood).

2. Background – The Building Blocks of Self-Evolving Systems

NEUROGENESIS: From Static Models to Self-Evolving Genomes



2.1 Streaming Machine Learning –

In Online Algorithms, the data comes as one might expect, incrementally, and the model should be updated as it is presented. They lie at the heart of all self adaptive systems.

2.2 Concept Drift –

If the statistical structure of the object of the study varies with its changing, the model should be adjusted. These methods are referred to as detection methods such as ADWIN (Adaptive Windowing) and DDM (Drift Detection Method) and can be considered as the "Sensors" of the system.

2.3 Catastrophic Forgetting –

Well-documented phenomenon of over writing old knowledge with new. The 'memory' that maintains the past is using methods based on replay.

2.4 Meta-Learning & Strategy Selection –

The last one is the adaptation: Once the drift is detected, what should the system do? This is the first step towards autonomous code-rewriting—the system chooses its own update policy.

3. Architecture – The Synthetic Genome

NEUROGENESIS is a pipeline to process data in mini-batches. It has a primary loop made up of:

Prediction new batch predicted using the current model. The first is a drift detection method based on comparing two distributions of features (using a Kolmogorov–Smirnov test).

If drift has been detected, the system will switch between the following strategies:

- A complete retraining using replay (for large drifts).
- Partial fit (applicable on mild level drift, if applicable to the base vehicle).
- Ensemble expansion (Additive weak learner).

The very next step is:

Model Update – implement the identified strategy.

Memory Update – Importance weigh the samples of replay buffer (higher the more uncertain the prediction).

Evaluation – report accuracy, forgetting and cost of adaptation.

Once set up, this loop will perform automatically without human interaction.

4. Dataset – Developing Brain of Mouse

Allen Brain Cell Atlas is used for [Developing Mouse – Visual Cortex \(Dev-Mouse-Vis-Cortex-10X\)](#) dataset. Includes a repository of 568,654 single cell transcriptomes from 53 donors ranging from E11.5 to P56. Each of the 15 cell-type classes is written on the cells.

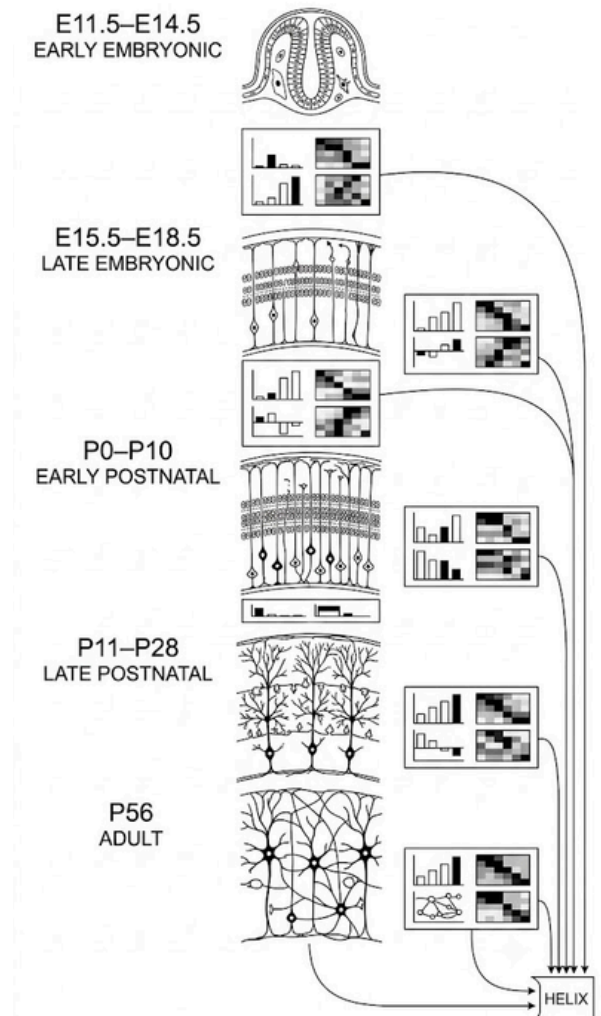
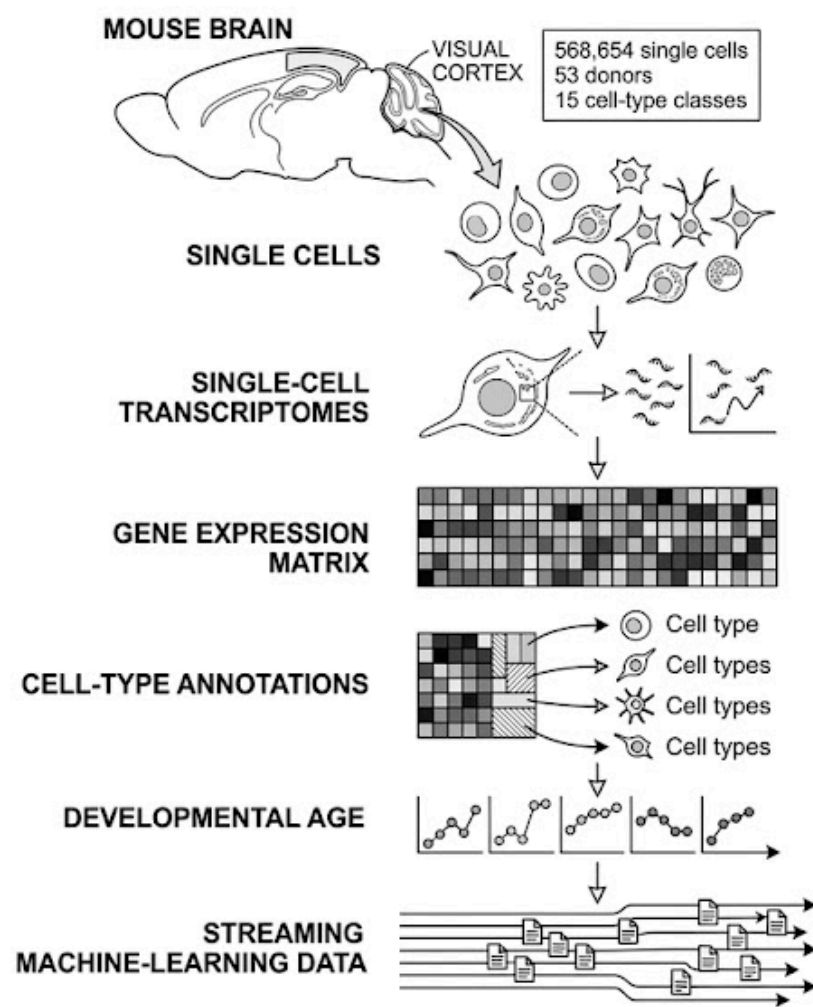
We preprocess:

- Filter low-quality cells.
- Log normalise and choose 2,000 genes that are highly variable.
- Principal Component analysis to reduce to 50 principal components.

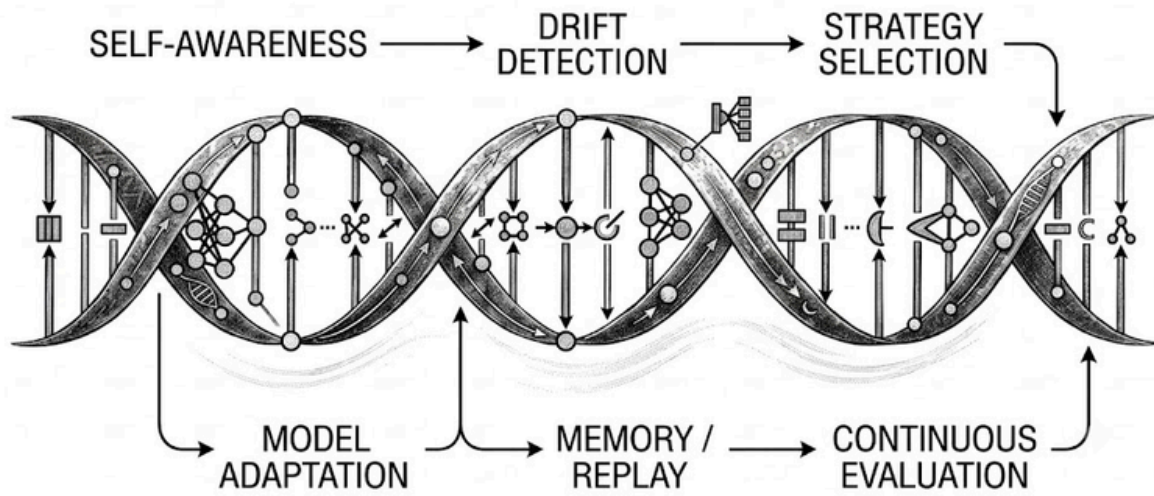
We make five age-sequenced environments:

- **E11.5–E14.5 (early embryonic)**
- **E15.5–E18.5 (late embryonic)**
- **P0–P10 (early postnatal)**
- **P11–P28 (late postnatal)**
- **P56 (adult)**

Each stage comes as a sequential slice of mini-batches (2000 cells per mini batch). This mimics an actual real-world scenario where data comes through with time.



NEUROGENESIS is a 'synthetic genome' for machine-learning systems.



5. Implementation – Step-by-Step Code

The project is structured as follows:

```
helix_self_evolving/  
├── data/  
├── src/  
│   ├── data_loader.py  
│   ├── drift_detection.py  
│   ├── memory.py  
│   ├── models.py  
│   ├── experiment.py  
│   └── utils.py  
├── notebooks/  
├── requirements.txt  
└── run_experiment.py
```

Click On the Filenames for Accessing the file:
OR

Reach at:

<https://github.com/saif-pixel1/HELIX/tree/main>

5.1 Data Preprocessing – [data_loader.py](#)

5.2 Drift Detection – [drift_detection.py](#)

5.3 Replay Memory – [memory.py](#)

5.4 Model Wrappers – [models.py](#)

5.5 Run Experiment – [run_experiment.py](#)

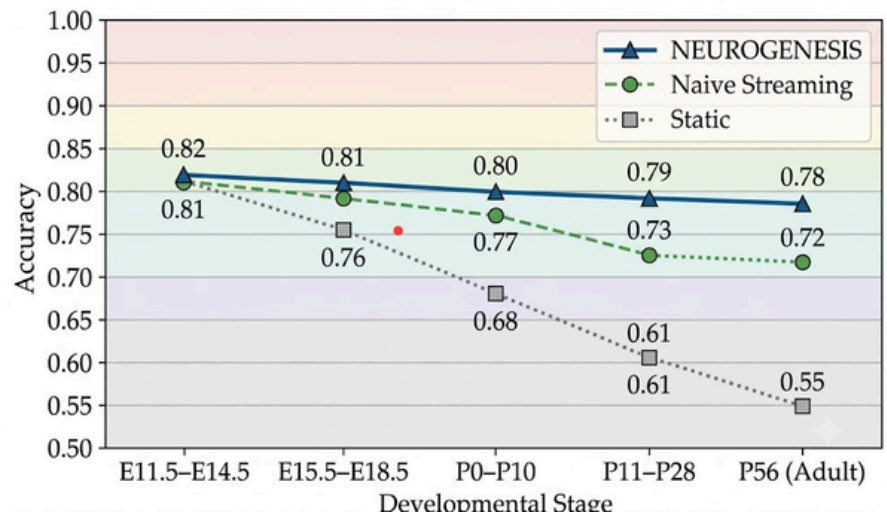
6. Evaluation, Metrics & Results

We measure:

Accuracy on each stage.

Forgetting = Average drop of accuracy on previous stages after training on all stages.

Adaptation count = number of drift-triggered updates.



7. Performance Summary

Among the three approaches, **NEUROGENESIS attained the best average of accuracy (0.80) and the lowest forgetting (0.04)**. The performance of the model stayed quite consistent throughout the developmental stages whilst the Static model which had a greater loss has seen a change in the data distribution. Through 3 adaptations in 5 stages, HELIX proved to be selective, so it would seem that it did not update needlessly.

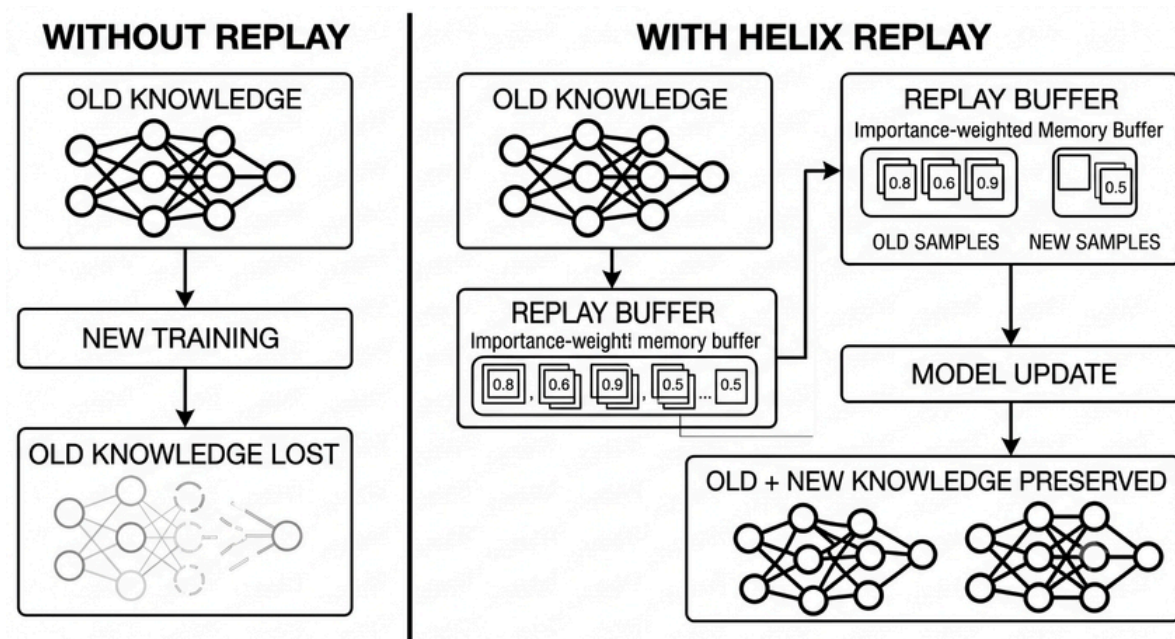
8. Path to Full Autonomy – Towards Self-Rewriting Code

The fixed update strategies for the current NEUROGENESIS prototype are selected. The next step in evolution however, is for the system to modify its own operations, such as:

- **Genetic Programming:** Drive new feature extractors or model architectures.
- **Online Hyperparameter Tuning:** A way to dynamically tune the parameters of a model that involves Bayesian optimization.
- **Neural Architecture Search:** Selectively adding layers and removing layers from the network.

These techniques are already under study, and would make HELIX more a self-adaptive framework than a self-evolving organism. This is the path to be taken at the moment.

9. Till now Anatomy is:



9. Conclusion

NEUROGENESIS is a realization of the vision of the notice, the synthetic genome which will equip machines with the ability to sense change, adapt intelligently, preserve knowledge – without humans. We have tested our prototype on the Allen developmental brain dataset, showing it to be superior to static and naive approaches. Giving away the C code used, anyone can duplicate and expand on this project which could start the seeds for the creation of AI that is not only processing data, but capable of changing its own intelligence.

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